Experiment 17

Thin Lenses and Virtual Images

INTRODUCTION

In this experiment, you will investigate virtual images and examine two important applications of these images: the magnifying glass and the telescope. One of the many claims to fame of Galileo Galilei is his work refining the design of a refracting optical telescope. Telescopes use combinations of lenses to produce a magnified image.

OBJECTIVES

In this experiment, you will

- Use ray diagrams to show the formation of virtual images by concave and convex lenses.
- Contrast characteristics of real and virtual images.
- Examine the conditions under which a convex lens acts like a magnifying glass.
- Use a combination of lenses to form a telescope.

MATERIALS

Vernier Dynamics System track

Vernier Optics Expansion Kit

PRE-LAB INVESTIGATION

Review the use of ray diagrams to help you to determine how and where light from a particular point on an object converges to form an image. You can get a conceptual understanding of the process of image formation by a lens using the "Geometric Optics" simulation available from the PhET web site.[1]

Place the light source assembly, 10 cm double convex lens, and viewing screen on the dynamics track, as shown in Figure 1. Turn on the light source and rotate the light source wheel until the number "4" is visible in the opening. Position the lens 10 cm from the light source. Can you find a place to put the screen where the image is in focus?

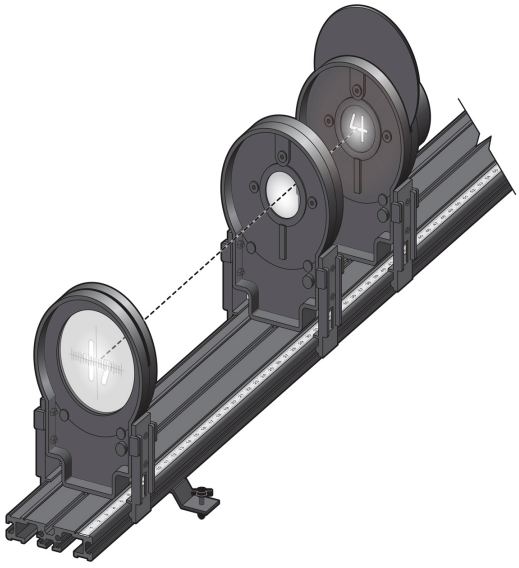


Figure 1

Remove the screen and hold the track so that the light passing through the lens projects onto a wall several meters away. Can you observe a focused image? Use either a ray diagram or the simulation to explain why a real image no longer forms when an object is placed at the focal point of the lens.

[1] http://phet.colorado.edu/en/simulation/geometric-optics

PART 1 INVESTIGATION OF VIRTUAL IMAGES

PROCEDURE

1. Return to the apparatus used in your Pre-Lab Investigation. Position the light source assembly and number "4" serving as the object slightly less than 10 cm from the lens. Remove the viewing screen from the track.

2. Position your eye approximately 30 cm from the lens holder and look through the lens at the illuminated "4." Record the appearance of the object.

3. Move your eye to one side of the lens so that you can directly see the illuminated "4." Compare the size and orientation of the image you can see through the lens to how the "4" appears when you view it directly. Record your observations.

4. Gradually move the lens closer to the object and observe the image by looking through the lens. What, if anything, about the image changes?

5. Replace the 10 cm convex lens with the 15 cm concave lens. Position the lens so that the illuminated "4" is at the focal point of the lens. Repeat Steps 2 and 3.

EVALUATION OF DATA

1. Sketch a ray diagram to show how the principal rays (parallel to the optical axis and through the center of the lens) from the top of the "4" diverge as they are refracted by the convex lens. Our eye interprets these diverging rays as *appearing* to originate from some point behind the lens. Using dashed lines, trace the diverging rays back until they converge behind the lens, then sketch the *virtual* image that is formed.

2. Describe the size, appearance, and location of the virtual image. Using the simulation, "Geometric Optics," move the object inside the focal point, then check the Virtual Image box to check your conclusions.

3. Use the thin lens equation to determine the location and size of the image formed when the "4" (2 cm high) is placed 8.0 cm from the 10 cm convex lens. What is the significance of the negative value for d_i ?

4. Repeat Step 1 using the concave lens. Note that the principal ray traveling parallel to the optical axis *diverges* as it passes through the lens. Its path, traced back behind the lens, passes through the focal point on the same side as the object. The ray passing through the center of the lens behaves the same as one passing through a convex lens. Using dashed lines, trace the diverging rays back until they converge behind the lens, then sketch the *virtual* image that is formed.

5. Describe the size, appearance and location of the virtual image. Use your text or a web resource to check your conclusions.

6. Use the thin lens equation to determine the location and size of the image formed when the "4" (2 cm high) is placed 10 cm from the 15 cm concave lens.

7. In what ways are these virtual images alike? How do they differ?

PART 2 COMBINATIONS OF LENSES

In your study of real images formed by convex lenses you are likely to have used a screen to help you locate and examine the images. It is important to note, however, that a screen is not necessary for a real image to form. In Figure 2, light rays emanating from a point on the object at left pass through the lens, converge at a point, then diverge again. If the eye of the observer is within the cone of rays diverging from that point in space, one can see an "aerial image" that appears to be floating in space. By placing a screen in the plane where the rays converge, it possible for observers to view the image from a variety of positions.

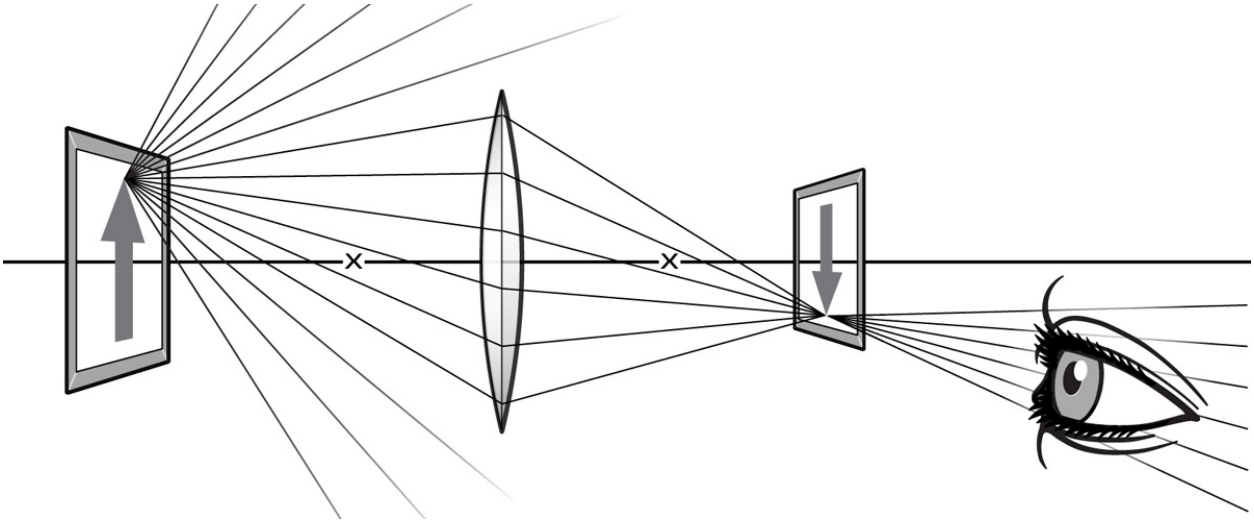


Figure 2 Aerial image

In this part of the experiment, you will examine aerial images and see how they can serve as the object for a second lens.

PROCEDURE

1. Position the light source assembly at the 10 cm mark on the track. Replace the 15 cm concave lens with the 20 cm convex lens. Position this lens at the 80 cm mark.

2. Position your eye near the end of the track so that it is aligned with both the illuminated "4" that serves as the object and the center of the lens. You should be able to observe the aerial image of the object. Describe the appearance of this image.

3. Is this aerial image real or virtual? How do you know?

4. Place the screen on the end of the track and move it toward the lens until the image comes into focus on the screen. Note the position of the screen.

5. Remove the screen and place the 10 cm convex lens at the 114 cm mark on the track. Position your eye near the end of the track so that it is aligned with both lenses and look at the illuminated "4." Describe the appearance of the image.

6. Is this image formed by the second lens real or virtual? How do you know?

7. Now, gradually move the 10 cm convex lens toward your eye. Note the effect this has on the appearance of the image.

EVALUATION OF DATA

1. By placing a second lens just beyond the plane where rays from the first lens converge you have made a telescope. Draw a ray diagram that shows the formation of the image by the first lens (objective) and how rays diverging from this image enter the second lens (eyepiece) and form an image.

2. Using the thin lens equation, determine the location and size of each image of the "4" when it is at the 10 cm mark, the 20 cm convex lens (objective) is at the 80 cm mark and the 10 cm lens (eyepiece) is placed at the 114 cm mark on the track. How many times larger is the final image when the eyepiece is at the 116 cm mark?

3. Use your text or web resource to determine the maximum magnification this telescope could provide. What would you have to change in order to improve the magnification of your telescope?

EXTENSION

Use a text or web resource to learn about the design of Galileo's original telescope. Contrast his design with the configuration of your telescope.

Aim your telescope at a distant object. Look through the eyepiece to examine the image of the object. Now, instead of the 10 cm convex lens, use the 15 cm concave lens as your eyepiece. Your instructor may help you determine where to place the eyepiece. Examine distant objects with this version of the telescope. Describe how this design affects your viewing experience. Why was Kepler's design considered superior to that of Galileo?